INCH-POUND
MIL-D-85866(AS)
15 October 1990

MILITARY SPECIFICATION

DELAY COMPOSITION, Z-1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture, acceptance, and delivery of one type of Z-1 delay composition (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-C-96	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-P-217	Potassium Perchlorate
MIL-B-550	Barium Chromate
MIL-C-10464	Cans, Hermetic Sealing, Metal, Light Gage, Tear-Strip Type
MIL-Z-11410	Zirconium-Nickel Alloy, Powdered

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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MIL-C-13739	Composition, Delay
MIL-P-22264	Powder, Ignition, Gasless, A-1A
MIL-B-82647	Bag, Conductive Plastic, Heat-Sealable, Flexible

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

423038	Replacement Sample Powder Tank, Mk 1 and Mods
LD 480089	Test Element and Output Charge Holder
1826685	Delay Body
1826740	Output Charge Holder, Loading Procedures

(Application for copies should be addressed to Commanding Officer, Naval Ordnance Station (Code 802), Louisville, KY 40214-5001.)

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

AD 1014	Operating Procedures for Pyrotechnic Delay Timing Apparatus

(Application for copies should be addressed to Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

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CODE OF FEDERAL REGULATIONS

29 CFR 1910.1200(g) Hazard Communication

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C516 Standard Specification for Vermiculite Loose Fill Thermal Insulation

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Material. The material shall be a homogeneous mixture of zirconium-nickel alloys, potassium perchlorate, and barium chromate in accordance with table I (see 4.5.1).

TABLE I. *Material.*

Ingredient	Specification	Weight (%)
Barium chromate	MIL-B-550, Grade A	60.0 ± 2.0
Potassium perchlorate	MIL-P-217, Grade A, Class 4	14.0 ± 2.0
Type I zirconium-nickel alloy 70/30 ratio	MIL-Z-11410	3.0 min to 26.0 max
Type II zirconium-nickel alloy 30/70 ratio	MIL-Z-11410	Remainder[1]

[1] Total zirconium and total nickel (elemental) to be determined per 4.5.1. Computed zirconium and nickel (totals) shall agree with chemical analysis ± 5.0%.

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3.3 Burning characteristics.

3.3.1 Burntime at 21 °C. The average burntime at 21 °C shall be as specified by the contracting activity (see 6.2) based on figures 1 and 2 (see 4.5.2).

3.3.2 Coefficient of variation of burntime at 21 °C. The coefficient of variation of average burntime at 21 °C shall not exceed 5 percent (see 4.5.3).

3.3.3 Percent variance of burntime at −54 °C. The percent variance of −54 °C burntime shall not exceed 15 percent of the average 21 °C burntime.

3.3.4 Coeffiicient of variation of burntime at −54 °C. The coefficient of variation of average burntime at −54 °C shall not exceed 5 percent (see 4.5.3).

3.4 Recertification. The delay mix shall be tested for recertification 1 year from date of manufacture (see 4.5.2 and 4.5.3).

3.5 Toxicological product formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the Material Safety Data Sheets (MSDS). Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.6 Material Safety Data Sheets (MSDS). The contracting activity shall be provided an MSDS at the time of contract award. The MSDS shall be prepared in accordance with the requirements of 29 CFR 1910.1200(g), Hazard Communication, and included with each shipment. OSHA Form 174 MSDS or a more effective contractor developed MSDS format may be used.

3.7 Workmanship. The delay composition shall be free of dirt, grease, and foreign matter (see 4.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for

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acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.2.1 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 3.5 and 6.7).

4.3 Lot. A lot of Z-1 delay composition shall consist of one manufacturer's batch. Each lot shall be less than 25 pounds. Each composition ingredient shall come from one ingredient lot.

4.4 First article inspection.

4.4.1 Sample size. The contractor shall submit a first article sample as designated by the contracting officer (see 6.4) for evaluation in accordance with provisions of 4.4.2. The sample shall be obtained from the first lot which has been produced by the contractor using the same production processes, procedures, and equipment used in production. The sample shall consist of 100 grams of delay compositon.

4.4.2 Inspection. The first article sample shall be subjected to all of the examinations and tests specified herein.

4.4.3 Sample rejection. Any sample failing to meet any of the requirements shall be cause for rejection. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.5 Quality conformance inspection. The composition shall be subjected to all inspections and tests herein for conformance. Sampling shall be in accordance with MIL-STD-105.

4.5.1 Material composition analysis. The composition shall be as specified in 3.2 when analyzed in accordance with MIL-C-13739.

4.5.2 Burntime test. The burntime test shall be as specified herein.

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4.5.2.1 Delay body preparation. The delay body shown in Drawing 1826685 shall be loaded in accordance with the instructions below with the exception that delay composition height shall be 0.5 inch.

a. Set tool post to 0.560 ± 0.002 inch.
b. Press approximately 3 equal increments of Z-1 composition in the delay body at 30,000 ± 500 psi. (Z-1 increments may be measured volumetrically.)
c. After the third increment is pressed, loosen the funnel cap and retighten. Repress this increment to improve compaction and minimize danger of powder fallout.
d. Remove delay body from tool. The surface of the delay column (pressed composition) shall be flush with the end of the delay body. Excess Z-1 powder may be removed by sliding the delay body over a soft cloth.
e. Press one ignition increment at opposite end of delay column. The ignition increment shall consist of approximately 160 milligrams of A-1A ignition powder conforming to MIL-P-22264 and shall be pressed at 30,000 ± 500 psi.

The tools required for the loading are specified in LD 480089. The output charge holder shall be loaded in accordance with Drawing 1826740.

4.5.2.2 Test apparatus preparation. The test assembly shall be prepared in accordance with AD 1014 or equivalent.

4.5.2.3 Temperature conditioning. The test adapters shall be conditioned continuously for at least 4 hours at the required temperatures, either 21 ± 4 °C (70 ± 7 °F) or −54 ± 4 °C (−65 ± 7 °F) before being subjected to functional test. The sample shall be divided equally for each temperature.

4.5.2.4 Functional test. The sample shall be tested within 2 minutes after removal from temperature conditioning. Fire the sample with an electric current of 2 amps, 28 volt dc power.

4.5.3 Computations. Sample average and standard deviation and coefficient of variation shall be cumputed as follows:

a. Take the sum of the individual measurements (ΣX).
b. Compute the average value by dividing the result of a by the number of measurements (N). Thus $\overline{X} = \Sigma X/N$.
c. Square each individual measurement (X^2).
d. Take the sum of the squared measurements (ΣX^2).
e. Multiply the result of d by the number of measurments (N). This gives $N(\Sigma X^2)$.
f. Square the result of a $(\Sigma X)^2$.
g. Subtract the result of f from the result of e.
h. Multiply the number of measurements (N) by the number of measurements less one (N-1).

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i. Divide the result of g by the result of h. This gives $[N(\Sigma X^2) - (\Sigma X)^2] / [(N)(N-1)]$.
j. Take the square root of the result of i. This gives standard deviation s. Thus,

$$s = \sqrt{\frac{N(\Sigma X^2) - (\Sigma X)^2}{N(N-1)}}$$

Compute the percent coefficient of variation by dividing $(100 \times s)$ by $\overline{X}$ for b.

4.5.4 Criteria of acceptability. The lot shall be rejected if the sample fails to comply with 3.3.2, 3.3.3, and 3.3.4 when tested in accordance wtih 4.5.2 and 4.5.3.

4.6 Inspection of packaging. An examination shall be made to determine if the packaging, packing, and marking comply with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the lot.

5. PACKAGING

5.1 Preservation.

5.1.1 Unit packaging. Place approximately 1200 grams of Z-1 delay composition into a double bag of 4-mil conductive plastic formed by inserting a 6 inch × 12 inch bag into another. The plastic bags shall conform to MIL-B-82647. Twist the top of the bag tightly, fold over once, and secure with two number (no.) 84 rubber bands.

5.1.2 Intermediate packaging. Place two double conductive bags in a metal ammunition can conforming to Drawing 423038. Expanded vermiculite conforming to ASTM C516 shall be used inside the intermediate container to ensure a tight pack. The net weight per can shall not be greater than 5 pounds of Z-1 delay composition.

5.1.2.1 Intermediate packaging, alternate no. 1. Place two double conductive bags in a can conforming to MIL-C-10464. Expanded vermiculite conforming to ASTM C516 shall be used to fill voids inside the intermediate container to ensure a tight pack prior to sealing in accordance with conventional canning techniques. This can is hermetically sealed. The net weight per can shall be not greater than 2500 grams or 5 pounds of Z-1 delay composition.

5.1.2.2 Intermediate packaging, alternate no. 2. Place two double conductive bags in a 1-gallon capacity, friction-top container conforming to PPP-C-96, type V, class 2. The cover shall be fixed by crimping at four or more points. Expanded vermiculite conforming to ASTM C516 shall be used on the inside of the intermediate container to ensure a tight pack. The net weight per container shall be not greater than 2500 grams or 5 pounds of Z-1 delay composition.

5.2 Packing. The packaged Z-1 delay composition shall be packed in a domestic type, wood cleated plywood box conforming to PPP-B-601 or in a class 1, style 2, 2½, 3, or 4 nailed wooden box conforming to PPP-B-621. Containers conforming to PPP-B-621 shall be lined with a material which will not cause

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nor produce a static electric discharge. Vermiculite conforming to ASTM C516 shall be used within the shipping box to provide separation of the intermediate containers and to ensure a tight pack. The gross weight of a shipping container shall be not greater than 50 pounds.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. In addition, the label on each shipping container shall contain the following information:

a. Z-1 delay composition
b. Name of manufacturer
c. Date of manufacture
d. Lot number in accordance with MIL-STD-1168
e. Burntime
f. Weight of composition, kilograms or pounds
g. Contract or purchase order number

5.5 Data cards. Data cards, in accordance with MIL-STD-1167, shall be retained by the contractor for each lot and made available for inspection by the contracting officer's representative. Each data card shall contain the following information:

Z-1 Delay Composition	Barium Chromate	Potassium Perchlorate
Manufacturer	Manufacturer	Manufacturer
Manufacturer's lot number	Manufacturer's lot number	Manufacturer's lot number
Original lot weight	Percent	Percent
Container net weight		
Burntime (average):	Zirconium-Nickel Alloy, Type I	Zirconium-Nickel Alloy, Type II
at 21 °C	Manufacturer	Manufacturer
at −54 °C	Manufacturer's lot number	Manufacturer's lot number
Coefficient of variation	Percent	Percent
at 21 °C		
at −54 °C		
Percent difference between burntimes at −54 and 21 °C		

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Z-1 delay composition is intended for use as a delay time determining element in ignition devices such as fuze explosive trains and cartridge actuated devices.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
c. Whether a first article sample is required (see 3.1, 4.4, and 6.4)
d. Burntime (see 3.3 and 4.5.2)
e. Quantity
f. Packaging and packing requirements (see section 5)
g. Special marking (see 5.3)
h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Consideration of data requirements.The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference Paragraph	*DID Number*	*DID Title*	*Suggested Tailoring*
4.5.1, 4.5.2, 4.5.3	DI-T-2072	Reports, test	Contractor format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractors current inventory, and the quantity of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 OSHA recommendation. The Occupational Safety and Health Administration (OSHA) recommends that the worker exposure to hexavalent chromium be controlled or eliminated. Barium chromate is an inferred carcinogen. The contractor is warned to make provisions for the protection of workers against toxic dust.

6.6 Advisory notes. The following notes are for information only:

6.6.1 Pilot mixture. Since the particle size distribution and chemical purity of the ingredients will vary from lot to lot, the burntime of delay compositions made from different ingredient lots may vary. A pilot mixture technique is therefore necessary to define the composition-burntime curve for the new components. A composition on the curve within the range of burntime of interest should be prepared and its burntime determined according to 4.5.2. Burntimes for the Z-1 delay composition are given in figures 1 and 2.

6.6.2 Drawing. Drawing NOS 23958-1 may be used in lieu of Drawing 1826685 for delay body loading.

6.6.3 Gas evolution. In design of units using Z-1 composition, consider that the mixture burns with some gas evolution. Venting or expansion chamber pressure relief is desirable to relieve pressure generated.

6.7 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of MSDSs. Additional Government mailing information is contained in Appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.8 Toxicological product formulations. Material safety data sheet requirements are applicable and should be specified in the contract as required by DOD FAR 23.303.

6.9 Subject term (key word) listing.

Zirconium-nickel alloys
Barium chromate
Potassium perchlorate

Preparing activity:
Navy - AS
(Project 1377-N987)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-85866(AS)	15 October 1990

3. DOCUMENT TITLE
DELAY COMPOSITION, Z-1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commanding Officer NAEC (SESD) Code 53	(1) Commercial (908) 323-1116	(2) AUTOVON 624-1116
c. ADDRESS (Include Zip Code) Lakehurst, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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